Quantifier rules yield provability facts

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-sequent-calculus/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the statement and proof use the tagenumerate/tagitem structure (tags prvEx/prvAll, FOL branch active as in the registered tableaux-unit sibling card), so both items render in full

Editorial note. The label prop:provability-quantifiers names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Left and right quantifier rules. Right-exists derives (exists x A(x)) on the right from A(t) on the right; Left-forall derives (forall x A(x)) on the left from A(t) on the left.. Item prvEx uses Right-exists on the initial sequent; item prvAll uses Left-forall.

Proposition 0.1. *1.* $A(t) \vdash \exists x\, A(x)$.

2. $\forall x\, A(x) \vdash A(t)$.

Proof. 1. The sequent $A(t) \Rightarrow \exists x\, A(x)$ is derivable:

$$\frac{A(t) A(t)}{A(t) \exists x\, A(x)}$$

2. The sequent $\forall x\, A(x) \Rightarrow A(t)$ is derivable:

$$\frac{A(t) A(t)}{\forall x\, A(x) A(t)}$$

□